REACHING POINTS AND CLOSURES IN THE NOETHERIAN LIMIT CASE

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Krull-Akizuki. Algebra, Krull-Akizuki lemma

Source reference. Original reference 02JQ: sources/stacks-chow/tags.txt, source lines 3345–3345. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01J5: sources/stacks-chow/tags.txt, source lines 2002–2002. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume f finite type and Y locally Noetherian. Let $y \in Y$ be a point in the closure of the image of f. Then there exists a commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a discrete valuation ring and K is its field of fractions mapping the closed point of $\mathrm{Spec}(A)$ to y. Moreover, we can assume that the image point of $\mathrm{Spec}(K) \to X$ is a generic point η of an irreducible component of X and that $K = \kappa(\eta)$.

Proof. By the non-Noetherian version of this lemma (Morphisms, Lemma 02JQ) there exists a point $x \in X$ such that $f(x)$ specializes to y. We may replace x by any point specializing to x, hence we may assume that x is a generic point of an irreducible component of X. This produces a ring map $\mathcal{O}_{Y,y} \to \kappa(x)$ (see Schemes, Section 01J5). Let $R \subset \kappa(x)$ be the image. Then R is Noetherian as a quotient of the Noetherian local ring $\mathcal{O}_{Y,y}$. On the other hand, the extension $\kappa(x)$ is a finitely generated extension of the fraction field of R as f is of finite type. Thus there exists a discrete valuation ring $A \subset \kappa(x)$ with fraction field $\kappa(x)$ dominating R by Algebra, Lemma 00PH. Then

$$\begin{array}{ccccccc} \mathrm{Spec}(\kappa(x)) & & & & & & X \\ \downarrow & & & & & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & \mathrm{Spec}(R) & \longrightarrow & \mathrm{Spec}(\mathcal{O}_{Y,y}) & \longrightarrow & Y \end{array}$$

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gives the desired diagram. □